

# PERFCTOID AND PRISMATIC METHODS

## 1. GETTING ACQUAINTED

**Exercise 1.1.** Let  $M \rightarrow N$  be a map  $p$ -complete modules if the reduction modulo  $p$  is injective/surjective/bijective, what can you say about the map? And if in addition  $M$  and  $N$  are  $p$ -torsion free?

**Exercise 1.2.** Let  $M$  be a  $p$ -complete module,  $p$ -torsion free. If  $M/pM$  is free, what can you say about  $M$ ?

**Exercise 1.3.** Let  $R \rightarrow S$  and  $R \rightarrow T$  map of perfect rings in characteristic  $p$ . Show that

$$S \otimes_R^L T \rightarrow S \otimes_R T$$

is an isomorphism.

**Exercise 1.4.** Show that a Noetherian perfect ring is a finite product of perfect fields.

**Exercise 1.5.** Let  $R$  be a normal domain such that it's fraction field is algebraically closed. Show that the Zariski and étale site of  $R$  are the same.

Now let  $X$  be an integral normal scheme with algebraically closed function field. What about the étale cohomology of  $X$ ?

**Exercise 1.6.** Let  $A$  be a  $\delta$ -ring such that  $\phi$  is injective. Show that then  $A$  is  $p$ -torsion free.

**Exercise 1.7.** Show that the  $p$ -completion of an absolutely integrally closed domain is perfectoid.

**Exercise 1.8.** Give examples of perfectoid rings, and show that they are perfectoid. Try to give at least one example of perfectoid ring with  $p$ -torsion, but not in characteristic  $p$ .

**Exercise 1.9.** Is the  $p$ -completion of  $\mathcal{O}_{\mathbb{C}_p} \otimes_{\mathbb{Z}_p} \mathcal{O}_{\mathbb{C}_p}$  perfectoid?

**Exercise 1.10.** Is there an initial perfectoid ring?